

H Measuring spread

MEASURES OF SPREAD SHOW THE RANGE OF DATA, AND ALSO GIVE MORE INFORMATION ABOUT THE DATA THAN AVERAGES ALONE.

Diagrams showing the measure of spread give the highest and lowest figures (the range) of the data and give information about how it is distributed.

Range and distribution

From tables or lists of data, diagrams can be created that show the ranges of different sets of data. This shows the distribution of the data, whether it is spread over a wide or narrow range.

Subject	Ed's results	Bella's results
Math	47	64
English	95	68
French	10	72
Geography	65	61
History	90	70
Physics	60	65
Chemistry	81	60
Biology	77	65

This table shows the marks of two students. Although their average (see pp.214–215) marks are the same (65.625), the ranges of their marks are very different.

lowest mark

highest mark

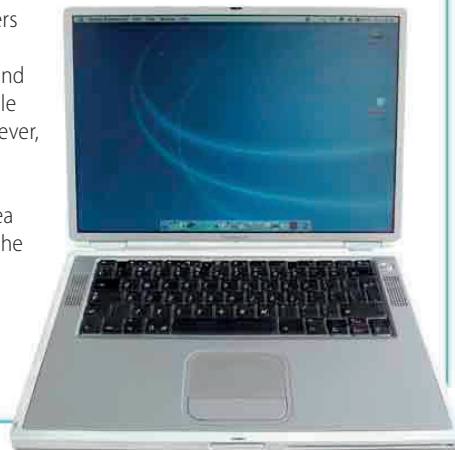
Ed: 10, 47, 60, 65, 77, 81, 90, 95

Bella: 60, 61, 64, 65, 65, 68, 70, 72

REAL WORLD

Broadband bandwidth

Internet service providers often give a maximum speed for their broadband connections, for example 20Mb per second. However, this information can be misleading. An average speed gives a better idea of what to expect, but the range and distribution of the data is the information really needed to get the full picture.



◁ Finding the range

To calculate the range of each student's marks, subtract the lowest figure from the highest in each set. Ed's lowest mark is 10, and highest 95, so his range is 85. Bella's lowest mark is 60, and highest 72, giving a range of 12.

